

GANESH SHARMA

Software Developer (Python/FastAPI)
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PROFILE SUMMARY

Python Backend Developer experienced in building REST APIs with FastAPI, PostgreSQL, SQLAlchemy, Alembic, Pydantic, and JWT-based authentication/authorization, with a strong focus on relational database design, asynchronous backend development, and modular API architecture. Also experienced in integrating LLM APIs into practical applications, with an emphasis on building reliable backend services around existing AI models.

TECHNICAL SKILLS

Languages: Python, C++

Back-End Engineering: FastAPI, Pydantic, SQLAlchemy, PostgreSQL, Alembic, RESTful APIs, Database Migrations

AI & Engineering: LLM APIs, LangChain, Prompt Engineering

Data/ML: Pandas, NumPy, Matplotlib, Scikit-Learn, ML Fundamentals

Dev Tools: Git, GitHub, Docker, VS Code, Agentic IDEs

EDUCATION

University of Mumbai

B.E. in Computer Science & Engineering (AI & ML)

Mumbai, India

2022–2026

PROJECTS

Multi-User Student & Academic Management Tool API | *FastAPI, Pydantic, SQLAlchemy, PostgreSQL, Alembic, JWT, RBAC*

- Architected and implemented a modular REST API for managing users, students, faculty, departments, roles, and academic records using FastAPI, asynchronous SQLAlchemy, and PostgreSQL, with relational models, foreign-key constraints, one-to-one/one-to-many relationships, validation schemas, filtering, and CRUD operations.
- Implemented secure authentication and role-based authorization using JWT bearer tokens, password hashing, protected /me user operations, and reusable authorization dependencies to restrict administrative and faculty operations based on user roles and permissions.
- Built a migration-driven, production-oriented persistence layer using Alembic for version-controlled PostgreSQL schema changes, including indexed/unique fields, database-generated identifiers, timestamps, enums, and relationship loading with SQLAlchemy selectinload, while structuring routes, models, schemas, authentication, and database dependencies into separate modules.

Full-Stack Blog Platform | *FastAPI, PostgreSQL, SQLAlchemy, Pydantic, Jinja2, CRUD*

- Architected a modular full-stack blog platform using FastAPI, PostgreSQL, and SQLAlchemy, isolating API routing, business logic, database models, and validation schemas to maximize system maintainability.
- Designed and implemented RESTful APIs with CRUD capabilities, executing robust data validation via Pydantic and robust token management via JWT-based authentication and role-based authorization.
- Engineered relational database schemas in PostgreSQL, leveraging SQLAlchemy ORM to manage complex entity relationships and optimize data persistence storage for efficient runtime queries.

AI Document Summarizer | *FastAPI, LangChain, LLM APIs, PyPDFLoader, TextLoader, WebBasedLoader*

- Engineered an AI-powered document summarization service using FastAPI, providing scalable RESTful endpoints for multi-format uploads, including PDF files, text documents, and web pages.
- Architected a modular processing pipeline for text extraction, prompt engineering, and LLM inference, utilizing LangChain to maintain a unified, easily extensible model interface.
- Implemented robust backend infrastructure, combining Pydantic request validation, comprehensive error handling, and structured API responses to ensure system reliability and seamless data delivery.

COURSES & CERTIFICATIONS

- Cursa's FastAPI Certification Course
- Harvard CS50's Introduction to Programming with Python (edX)
- Google's AI Essential Course — Coursera

RESEARCH & PUBLICATIONS

MITRA AI Chatbot Assistant: Using 8-Bit Quantized Large Language Models on Consumer-Grade GPUs

Published in *International Journal for Research in Applied Science and Engineering Technology (IJRASET)*

Volume 14, Issue III, pp. 5059–5065, ISSN: 2321-9653.